Kumar Gaurav Mishra

22dr0113@iitism.ac.in | LinkedIn | ResearchGate | Google Scholar

Professional Summary

Fourth-year PhD candidate in Environmental Science and Engineering at IIT (ISM) Dhanbad, working on atmospheric chemistry with a focus on fine particulate matter. My research has progressively covered the chemical, optical, isotopic, and thermodynamic properties of $PM_{2.5}$ and PM_{10} in a coal-industrial environment, combining field measurements with receptor modelling, back-trajectory analysis (HYSPLIT), and aerosol thermodynamic modelling (ISORROPIA-II/HETP). Expected thesis submission: December 2026; defence: March 2027.

Education

PhD in Environmental Science and Engineering — Aug 2022 – Present
Indian Institute of Technology (ISM) Dhanbad, India — GPA: 8.0/10.0

Supervisor: Professor Saifi Izhar

MSc in Environmental Science — Dec 2020 – May 2022
Central University of South Bihar, Gaya, India — GPA: 8.42/10.0

Mentor: Professor Pradhan Parth Sathi

BSc in Botany (Honours) — Aug 2017 – Dec 2020
Gaya College, Magadh University, Gaya, India — 70.74%

National Qualifications

UGC-NET, Environmental Sciences (2024) | GATE, Ecology and Evolution (2022)

Research Experience

- **PhD Research — Aerosol Lab, IIT (ISM) Dhanbad** — Aug 2022 – Present

  Year-long field campaign (March 2023 – March 2024, $N = 131$ paired samples) characterising $PM_{2.5}$ and PM_{10} in Dhanbad, designed for source apportionment; the same dataset underpins the thermodynamic work. Multiple sampling campaigns run in parallel (ambient and primary emission-source sampling). Work has spanned:

  - *Chemical properties:* Water-soluble ionic composition (IC), OC/EC (thermal-optical reflectance), metals (ICP-AES, ED-XRF)
  - *Optical properties:* Light absorption and source-resolved optical characterisation of carbonaceous aerosol
  - *Isotopic properties:* Carbon isotope ($\delta^{13}C$) analysis for source attribution of carbonaceous fractions
  - *Thermodynamic properties:* Aerosol pH, aerosol liquid water content (ALWC), and gas–particle partitioning using ISORROPIA-II and HETP v1.0; counterfactual simulations to quantify non-volatile cation effects on nitrate stability
  - *Source apportionment:* Emission source profiling; PMF and ME-2 receptor modelling (in progress)
  - *Transport analysis:* Back-trajectory cluster analysis using HYSPLIT/TrajStat to link air mass origins to aerosol composition

- **Laboratory Setup — Aerosol Lab, IIT (ISM) Dhanbad** — 2022 – Present

  Designed and commissioned the Aerosol Lab (layout, instrument siting, power and UPS load planning, procurement); manage day-to-day operations and consumables.

Technical Skills

- **Programming & Data Analysis**
  Python (primary): pandas, NumPy, matplotlib, seaborn, scikit-learn, statsmodels, SciPy
  Spatial analysis: geopandas, QGIS
  R (specialised): openair, ggplot2

- **Atmospheric Modelling**
  ISORROPIA-II / HETP v1.0 (aerosol thermodynamics)
  PMF / ME-2 (receptor modelling)
  HYSPLIT / TrajStat (back-trajectory and cluster analysis)
- **Laboratory & Instrumentation**
  Ion chromatography (IC), thermal-optical reflectance (TOR), ICP-AES, ED-XRF, stable carbon isotope analysis ($\delta^{13}C$)
- **Document Preparation**
  LaTeX (manuscripts, reports, presentations), Microsoft Office

Publications

Peer-reviewed

[1] **Mishra, K. G.**, Izhar, S., Paul, D., Qadri, A. M., Gupta, T. (2025). Insights on optical absorption and isotopic properties of carbonaceous aerosol in $PM_{2.5}$ and PM_{10} from different emission sources. *Environmental Pollution*, 382, 126558. doi:10.1016/j.envpol.2025.126558

[2] **Mishra, K. G.**, Prakashmani, Gupta, T., Izhar, S. (2024). Source profile of $PM_{2.5}$ emissions from different primary sources in the coal capital city Dhanbad, India. *Atmospheric Environment: X*, 21, 100235. doi:10.1016/j.aeaoa.2024.100235

Under review

[3] **Mishra, K. G.**, et al. Non-volatile cation buffering controls seasonal aerosol acidity, water content and nitrate partitioning in the coal-industrial eastern Indo-Gangetic Plain. *Environment International* (under review; ENVINT-D-26-04594).

In preparation

- Source apportionment of $PM_{2.5}$ and PM_{10} (PMF/ME-2) using locally measured Dhanbad source profiles.
- Hygroscopicity (κ) of $PM_{2.5}$ and PM_{10} in a coal-industrial environment.

Conference Presentations

- **Mishra, K. G.**, Izhar, S. (2026). "Nitrate Limitation and Influence of Air Mass Origins on Aerosol pH and ALWC in a Coal Mining City." Oral presentation (Abstract #822), iCACGP-IGAC 2026 Joint Science Conference, Heraklion, Crete, Greece.
- **Mishra, K. G.**, Izhar, S. (2026). "Influence of Air Mass on Atmospheric Thermodynamics and Secondary Aerosol Formation in a Complex Industrial Environment." Accepted for oral presentation (ID 37364, Session A-21); abstract published, not presented. 12th International Aerosol Conference (IAC 2026), Xi'an, China.
- **Mishra, K. G.**, Izhar, S. (2025). "Assessing the Influence of Meteorological Variables on $PM_{2.5}$ Concentrations under Sparse Observation Regimes: Application of Nonparametric Statistical and Machine Learning Methods." Poster presentation, India Clean Air Summit (ICAS 2025), CSTEP, Bangalore, India.
- **Mishra, K. G.**, Izhar, S., Paul, D., Qadri, A. M., Gupta, T. (2024). "Optical and Isotopic Source Profile in Size Segregated Traffic and Combustion Sources in Dhanbad." (24IA11230). Poster presentation, IASTA 2024: Aerosol Impacts to Air Quality, Human Health & Climate Change, Dehradun, India.
- Izhar, S., **Mishra, K. G.**, Paul, D., Gupta, T. (2024). "Effect of Emission Sources and Size on Carbonaceous Aerosol Optical and Isotopic Characteristics." Oral presentation, 13th Asian Aerosol Conference (AAC 2024), Kuching, Sarawak, Malaysia.
- **Mishra, K. G.**, Prakashmani, Gupta, T., Izhar, S. (2023). "Source Profile of Carbonaceous $PM_{2.5}$ Emissions from Different Primary Sources in the Coal Capital City Dhanbad, India." (T3017). Poster presentation, IASTA National Conference 2023, Mumbai, India.

Grants

- Travel grant, India Clean Air Summit (ICAS 2025), CSTEP, Bangalore 2025
- Travel grant, CAPHER workshop on Air Pollution and Health Research, AIIMS Guwahati 2025

Professional Service

- Peer reviewer, *Atmospheric Environment: X*

Teaching Experience

Teaching Assistantships

- Winter 2025-26: Environmental Computational Lab (NESC511)
- Monsoon 2025-26: Cleaner Energy (ESD405)
- Winter 2024-25: Air Pollution (NESC103), Air and Noise Pollution Lab (NESC104), Research Methodology (NESC595 & NESC596), Environmental Computational Lab (NESC511)
- Monsoon 2024-25: Aerosols in the Atmosphere (ESD511), Cleaner Energy (ESD405), Air and Noise Pollution Lab (ESC252)
- Winter 2023-24: Research Methodology (ESC526)
- Monsoon 2023-24: Aerosols in the Atmosphere (ESD511), Cleaner Energy (ESD405)

Guest Lectures

- Resource person, three-day Management Development Program on Python-Based Data Analytics for BCCL Executives; honorarium at Assistant Professor rate — Feb 2025

Training and Professional Development

W1. *International Workshop on Air Quality Monitoring and Data Analysis.* IIT (ISM) Dhanbad — Feb 2026

W2. *Fundamentals of Aerosol Science 2025.* The Aerosol Society, UK — Feb 2026

W3. *Introduction to REACH Reduced-Complexity Air Quality Models.* ICAS–CSTEP, Bangalore — Aug 2025

W4. *Designing and Implementing Air Pollution and Health Research Studies.* CAPHER & AIIMS Guwahati, Guwahati — Aug 2025

W5. *5th IPSL Summer School: Atmospheric Chemistry and Air Quality.* Sorbonne University, Paris, France (Remote) — Jun–Jul 2025

W6. *Executive Development Program: Integrated Low-Cost Sensors and Automation Technologies for Air Quality Monitoring in Mining Industry.* IIT (ISM) Dhanbad — Feb 2025

W7. *CAPHER–IASTA Pre-Conference Workshop: Writing Successful Research Proposals for Air Pollution and Health.* Doon University — Dec 2024

W8. *National Training on Advanced Aerosol Instrumentation.* IITM Pune & IASTA, Pune — Sep–Oct 2024

W9. *NASA ARSET: Atmospheric Composition Ground Networks Supporting Air Quality and Climate Applications.* Remote — Aug 2024

W10. *Indo–UK Workshop: Addressing Indoor Air Quality Challenges in Dhanbad.* IIT (ISM) Dhanbad — May 2024

W11. *Aerosol Science Intensive Course.* IIT Delhi & University of Helsinki, Sonipat — Feb 2024

W12. *NASA ARSET: Fundamentals of Machine Learning for Earth Science.* Remote — Apr–May 2023

Leadership

President, Hostel Executive Committee — Dec 2024 – May 2025
Emerald Hostel, IIT (ISM) Dhanbad

Led administrative operations, organised institute-level events, and streamlined hostel processes.